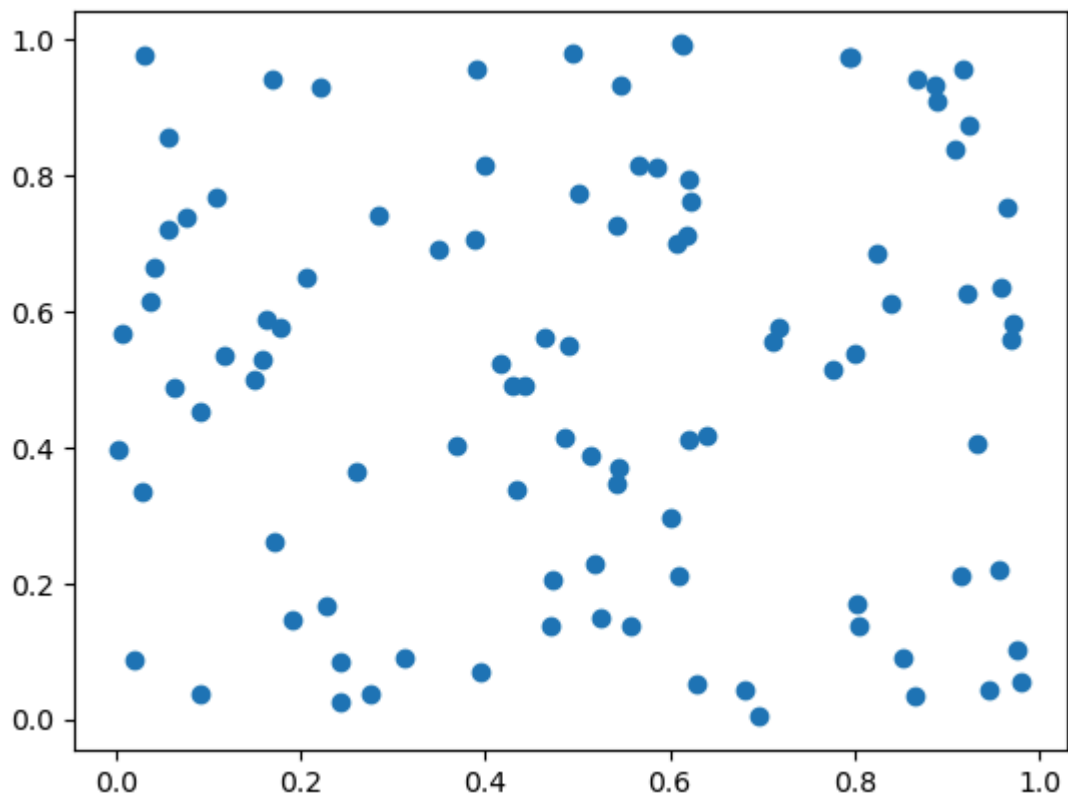


```
In [1]: import numpy as np
        from sklearn.cluster import KMeans
        from sklearn.metrics import silhouette_score
        import matplotlib.pyplot as plt
```

```
In [2]: X = np.random.rand(100,2)
```

```
In [3]: plt.scatter(X[:,0], X[:,1])
        plt.show
```

```
Out[3]: <function matplotlib.pyplot.show(close=None, block=None)>
```



```
In [5]: kmeans = KMeans(n_clusters=3, random_state=0)
        kmeans.fit(X)
```

C:\Users\Shabnam kumari\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.
warnings.warn(

```
Out[5]: ▼ KMeans ⓘ ?
        ► Parameters
```

```
In [6]: centroids = kmeans.cluster_centers_
        labels = kmeans.labels_
```

```
In [9]: print(centroids)
        print(labels)
```

```

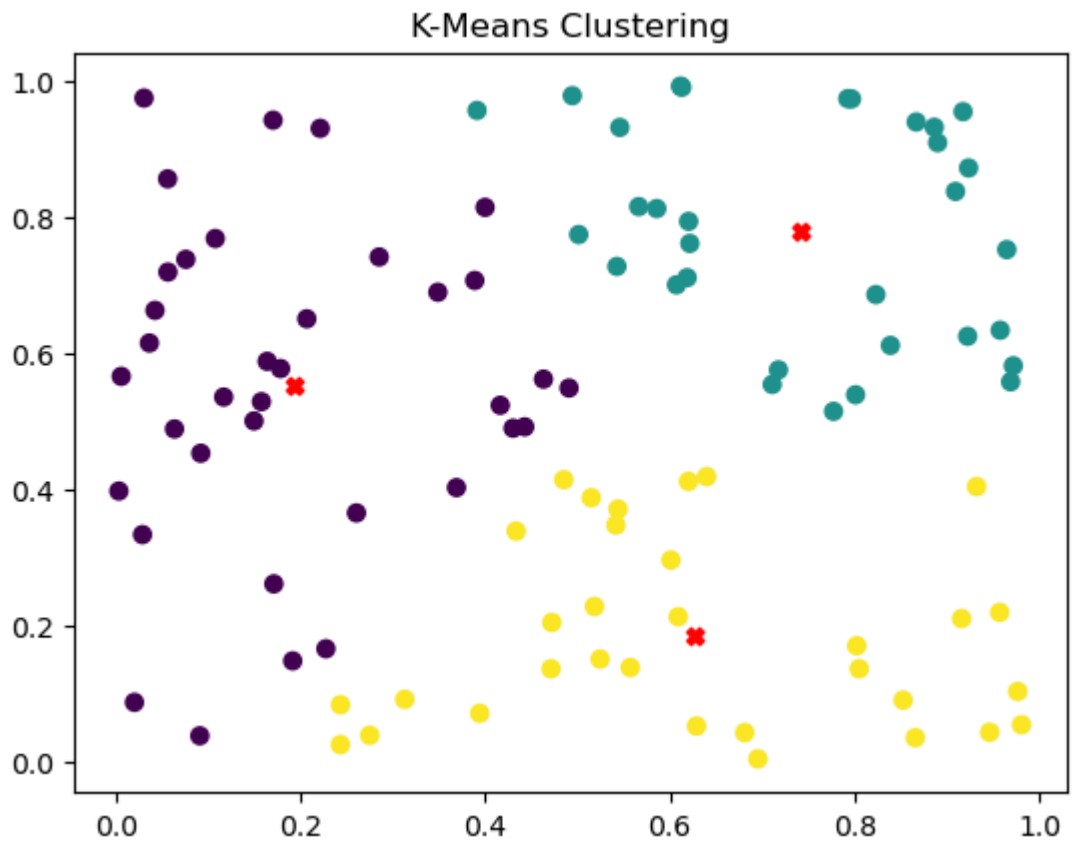
[[0.19357709 0.55189729]
 [0.74268535 0.78078261]
 [0.62655787 0.18499389]]
[0 0 2 1 0 2 0 0 0 2 2 0 0 1 2 1 0 2 2 0 2 1 1 2 2 0 2 1 1 0 0 1 0 2 0 1 2
 0 1 1 1 2 2 0 0 0 0 1 0 2 1 1 0 0 2 2 2 0 1 1 2 1 1 0 1 1 0 2 0 1 2 0 1 2
 2 2 2 1 0 1 0 1 0 2 0 2 1 2 2 2 1 1 0 1 1 1 2 0 0 0]

```

```
In [10]: print(f"Silhouette Score: {silhouette_score(X, labels):.4f}")
```

Silhouette Score: 0.4060

```
In [13]: plt.scatter(X[:,0],X[:,1],c=labels)
plt.scatter(centroids[:,0], centroids[:,1], marker='x', color='red')
plt.title('K-Means Clustering')
plt.show()
```



```
In [ ]:
```